Department of Information Technology

Academic Year 2024 - 2025 (Even Semester)

Degree, Semester & Branch: IV Semester B.Tech. IT
Course Code & Title: CS3452 & Theory of Computation
Name of the Faculty member (s): Dr.V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic:** Unit IV / Pumping lemma for CFL
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO13
- **Activity Chosen:** Collaborative Learning
- **Justification:**
  In unit IV, The **Pumping Lemma for Context-Free Languages (CFL)** is a theoretical and abstract concept that requires higher-order thinking and logical reasoning to understand and apply. Many students often find it challenging due to its formal structure and the need to construct or disprove proofs for given languages.
  To address this challenge, Collaborative Learning was chosen.

- **Time Allotted for the Activity:** 10 Minutes

- **Details of the Implementation:**
  - The topic pumping lemma for CFL was taught in the previous classes. The time allotted for the activity is 10 Minutes.
  - A team of five members assigned one problem statement as given in the topic pumping lemma for CFL as shown in Figure2.
  - The team members discussed and completed the conversion as shown in Figure 1.
  - Sample answer for the solutions is as shown in Figure 3. The student Divya Sri J S presented as shown in Figure 4.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO4	2	3	3	2	2	1	2	2	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO4	2	3	3	2	2	1	2	2	2	2	2
Justification for correlation	Apply basic Knowledge and fundamentals of Theory of Computation for Pumping lemma for CFL.	The students will be able to identify the need for analysis of complex engineering problems	Able to design the process flow for Pumping lemma for CFL	Able to analyze the data for Pumping lemma for CFL	The students will be able to follow ethics for solving problems	The students able to function as a team for Pumping lemma for CFL	The students will be able to communicate one another and can able to present	The students able to learn and implement the concept in Natural Language Processing	The students able to apply mathematical and computing skills.	The students can Exhibit knowledge of languages in the field of Cloud Computing to develop IT solutions	The students able to exhibit knowledge in Languages to develop a quality product.

- **Images / Screenshot of the practice:**

Figure 1: Collaboration Activity

1. Lanbncn = $\{a^n b^n c^n \mid n \geq 0\}$ is not a CFL.
2. Find out whether L=$\{x^n y n^z n \mid n>=1\}$ is context free or not.
3. L a=c∧b=d = $\{a^i b^j c^i d^j \mid i, j \geq 0\}$ is not a CFL.
4. E = $\{ww \mid w \in \{0, 1\}*\}$ is not a CFL.
5. L1 =$\{ a^i b^i c^i \mid I,j>=0\}$ is not a CFL.
6. L2= $\{ a^i b^j c^i d^j \mid i>=1$ and $J>=1 \}$ is not a CFL.
7. Show that L=$\{a^p$/p is prime$\}$ not context free.
8. Discuss the closure properties of CFLS.
9. Show that language $\{a^i b^j c^i d^j/i>=1$, and $j>=1\}$ is not context-free.
10. Show that $(0^n 1^n 2^n/n>=1\}$ is not a Context free language.
11. Prove that the language L=$\{a^n b^n c^n \mid n>=1\}$ is not context free using pumping lemma.
12. Prove that $L = \{a^n \mid n \text{ is a Perfect square}\}$ is not context free.

Figure 2 Problems given to Students

2) Prove $L = \{a^i b^i c^i \mid i,j > 0\}$ is not regular

$L = \{abc\, aaabbc, \ldots\}$

$Z = aabbcc \qquad n = 26$

i) $|uv| <= n$ ii) $|v| >= 1$. iii) $uvw \Rightarrow aaabbcccc \notin L$

$4 <= b$ $\qquad 3 >= 1$

The ~~string~~ doesn't belongs to the language. The language is not regular.

Prove $L = \{a^i, b^i c^i \mid i,j > 0\}$ is not regular.

$L = \{abc, aabbcc \ldots\}$.

$Z = aabbcc$

1) $|uv| <= n, \Rightarrow 4 <= 6$.

2) $|v| >= 1 \Rightarrow 3 >= 1$

3) $uv^iw \Rightarrow aaabbcccc \notin L$.

$\therefore$ The string does not belong to the language. and not Regular.

Figure 3 Solution given by the Students Ms.R. Priyadharshini and Ms.R. Sibiah

Figure 4 Presentation by Divyasri J S

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***
  - Working in a group helped the students to understand the steps involved in applying the Pumping Lemma more clearly. Discussing different approaches made the concept easier to grasp.
  - Through discussion and feedback, the students identified the mistake. It was a good learning experience.
  - The students reported better **concept clarity**, improved **communication skills**, and increased **engagement** in the learning process.
  - The activity was very engaging and interactive. Instead of passively listening, we were thinking and solving problems together.
- ***Benefit of the practice:***
  - The student said they were hesitant to solve theoretical problems before, but now they feel more confident after collaborating with others.
  - Initially confused about how to break the string into parts for proof, but the team mates explained it well. The students learned a lot from their explanations.
  - Most students **appreciated the collaborative format**, stating that it helped them **understand a difficult topic more effectively**.
- ***Challenges faced in implementation:***
  - It would have been helpful if the students had gone through one more example as a class before starting the group activity.
  - The time felt a bit short for solving complex problems. Slightly extending the duration would help.

References:

- https://teaching.cornell.edu/teaching-resources/active-collaborative-learning/collaborative-learning
- https://www.nureva.com/blog/education/15-active-learning-activities-to-energize-your-next-college-class

Signature of Faculty Member **HOD**